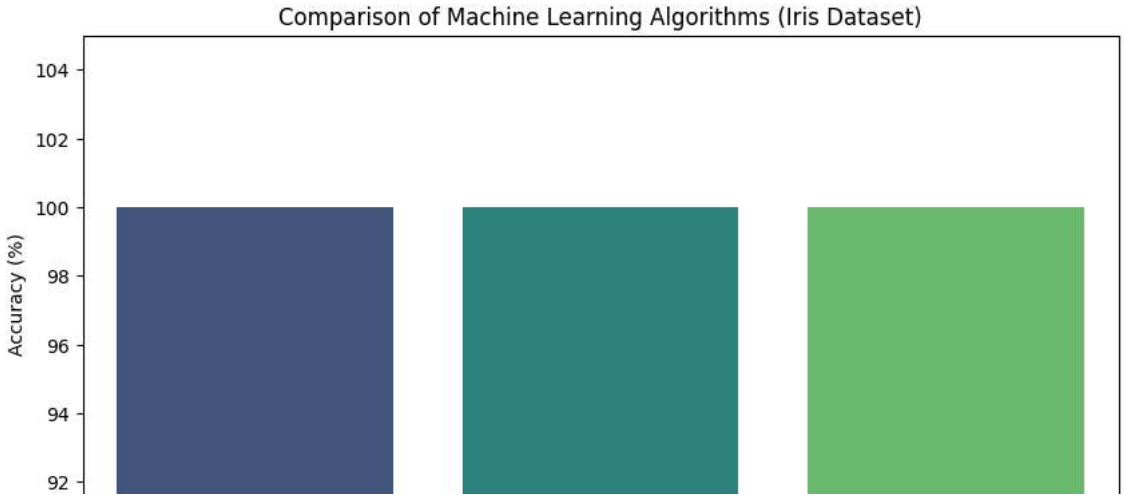
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

# 1. Load the Dataset
iris = load_iris()
X = iris.data
y = iris.target

# 2. Split into Training (80%) and Testing (20%)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 3. Define the 3 Algorithms
models = {
    "Logistic Regression": LogisticRegression(max_iter=200),
    "Decision Tree": DecisionTreeClassifier(),
    "K-Nearest Neighbors": KNeighborsClassifier(n_neighbors=5)
}

# 4. Train and Evaluate
results = []
for name, model in models.items():
    model.fit(X_train, y_train)
    predictions = model.predict(X_test)
    accuracy = accuracy_score(y_test, predictions)
    results.append({"Algorithm": name, "Accuracy (%)": round(accuracy * 100, 2)})

# 5. Display the Comparison Table
comparison_df = pd.DataFrame(results)
print("--- ALGORITHM COMPARISON TABLE ---")
print(comparison_df.to_string(index=False))

# 6. Create a Visualization for your lecture
plt.figure(figsize=(10, 5))
sns.barplot(x='Algorithm', y='Accuracy (%)', data=comparison_df, palette='viridis')
plt.title('Comparison of Machine Learning Algorithms (Iris Dataset)')
plt.ylim(90, 105) # Zoom in to see differences
plt.show()
```

```
--- ALGORITHM COMPARISON TABLE ---
          Algorithm  Accuracy (%)
Logistic Regression         100.0
      Decision Tree         100.0
K-Nearest Neighbors         100.0
/tmp/ipykernel_767/611415044.py:41: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` an

  sns.barplot(x='Algorithm', y='Accuracy (%)', data=comparison_df, palette='viridis')
```